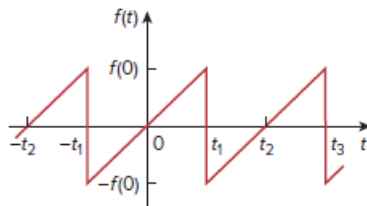


Problem

Using Fig. 17.61, design a problem to help other students better understand finding the Fourier series of a periodic wave shape.

Figure 17.61:



Step-by-step solution

Step 1 of 1

Using Fig. 17.61, design a problem to help other students better understand finding the Fourier series of a periodic wave shape.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Find the Fourier series of the function shown in Fig. 17.61.

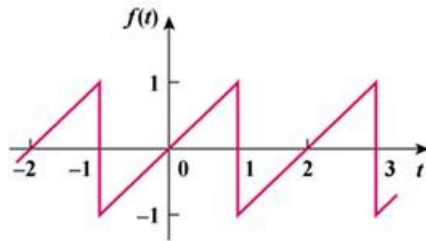


Figure 17.61

Solution

$f(t)$ is an odd function.

$$f(t) = t, -1 < t < 1$$

$$a_0 = 0 = a_n, T = 2, \omega_0 = 2\pi/T = \pi$$

$$b_n = \frac{4}{T} \int_0^{T/2} f(t) \sin(n\omega_0 t) dt = \frac{4}{2} \int_0^1 t \sin(n\pi t) dt$$

$$= \frac{2}{n^2 \pi^2} [\sin(n\pi t) - n\pi t \cos(n\pi t)]_0^1$$

$$= -[2/(n\pi)] \cos(n\pi) = 2(-1)^{n+1}/(n\pi)$$

$$f(t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(n\pi t)$$